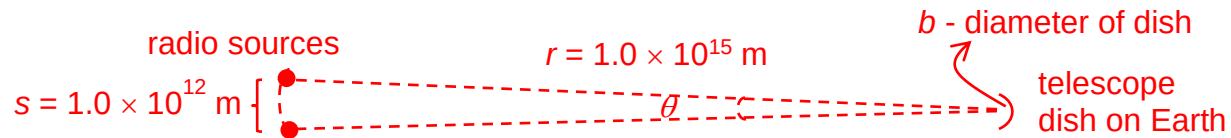


- 18** Two sources of radio waves are at a distance of 1.0×10^{15} m from Earth. The sources are separated by 1.0×10^{12} m and emit radio waves of wavelength 0.030 m.

What is the estimate for the diameter of a dish of a radio telescope on Earth that will just resolve the two sources?

- | | | | | | | | |
|----------|------------------------|----------|------------------------|----------|--------|----------|------|
| A | 3.0×10^{-5} m | B | 3.0×10^{-4} m | C | 0.03 m | D | 30 m |
|----------|------------------------|----------|------------------------|----------|--------|----------|------|

Answer: D



Let s be distance between the 2 radio sources,
 r be distance between radio sources and Earth,
 θ be minimum angular separation subtended by the two radio sources at the telescope dish on Earth,
 b be diameter of telescope dish on Earth.

Since the radio sources are at a very far distance from the Earth,

$$s \approx r \theta$$

$$1.0 \times 10^{12} = (1.0 \times 10^{15}) \theta$$

$$\theta = 1.0 \times 10^{-3} \text{ rad}$$

For the telescope dish to just resolve the two sources,

$$\theta = \frac{\lambda}{b} \Rightarrow b = \frac{\lambda}{\theta} = \frac{0.030}{1 \times 10^{-3}} = 30 \text{ m}$$

- 19** Which of the following statements describing a gravitational and an electric field is correct?